Data-Driven Apartment Rent Prediction in U.S. Urban Markets

Parastoo Hashemi Alvar
Politecnico di Torino
s339438
s339438@studenti.polito.it

***Abstract*—We develop a machine learning pipeline for U.S. apartment rent prediction that addresses data quality challenges through targeted preprocessing (log transforms, median imputation, Bayesian-smoothed location encoding) and feature engineering. Evaluating five regression algorithms with hyperparameter tuning, our optimized Random Forest [2] model achieves $186.2 MAE (23% better than baselines), with feature importance revealing location (38%) and unit size (22%) as primary price drivers. The solution balances accuracy with interpretability, offering actionable insights for housing market participants.**

I. Problem Overview

Accurately forecasting monthly apartment rents is vital for tenants, landlords, investors, and urban planners to make informed decisions and ensure fair market practices. In this project, we leverage a rich dataset of 99,487 U.S. apartment listings—partitioned into a development set of 79,589 labeled entries and an evaluation set of 19,898 unlabeled listings—for training and validating predictive models that minimize Mean Absolute Error (MAE) on held-out data [1]. The dataset spans numeric attributes (e.g., square footage, room counts), binary indicators (e.g., pets allowed, photo present, amenity flags), and location information (city and state).

Figure 1 illustrates the right-skewed nature of the rental price distribution, where most listings fall between $1 000 and $2 000, but a long tail stretches toward luxury properties priced up to $52 000. This motivated a log transformation of the target variable to reduce heteroscedasticity and stabilize model training.

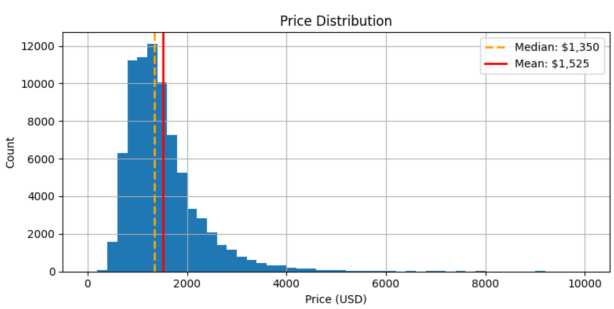

Fig. 1. Histogram of monthly rent prices in the development set, showing a heavy right-skew and a long tail of high-end listings.

Tables I and II summarize the key descriptive statistics. Notably, some variables present substantial missingness: 55.6% of listings lack pet policy data, and 16.1% have missing amenity information. Furthermore, high-cardinality location variables like `cityname` (2 806 unique cities) and `state` (51 unique entries) required special encoding strategies to avoid overfitting.

TABLE I
Descriptive Statistics for Numeric Variables

Variable	Missing	Mean	Median	Std	Min	Max
bathrooms	51	1.44	1.00	0.55	1.00	8.50
bedrooms	89	1.73	2.00	0.75	0.00	9.00
price	0	1525.03	1350.00	892.08	100.00	52500
square_feet	0	955.42	900.00	391.49	101.00	40000
latitude	23	36.95	37.23	4.60	19.57	64.83
longitude	23	-91.57	-84.56	15.81	-159.37	-68.78

TABLE II
Summary of Non-Numeric Variables

Variable	Missing	% Missing	Values	Unique	Type
category	0	0.00	79589	7	object
title	0	0.00	79589	48382	object
body	0	0.00	79589	76109	object
amenities	12804	16.09	66785	8929	object
currency	0	0.00	79589	1	object
fee	0	0.00	79589	2	object
has_photo	0	0.00	79589	3	object
pets_allowed	44244	55.59	35345	5	object
price_type	0	0.00	79589	1	object
address	73226	92.01	6363	6239	object
cityname	234	0.29	79355	2806	object
state	234	0.29	79355	51	object
source	0	0.00	79589	22	object

A correlation analysis (see Fig. 2) reveals strong multicollinearity between structural features—particularly between `square_feet` and `bathrooms` ($r = 0.72$)—and a moderate correlation between price and unit size ($r = 0.39$). Interestingly, the relationship between rent and room counts (bedrooms, bathrooms) is weaker, suggesting that physical size is a more direct driver of rental pricing than just the number of rooms.

Key objectives of this study are:

- Clean and preprocess heterogeneous rental listing data.
- Engineer interpretable features (room ratios, amenity flags, smoothed geographic encodings).
- Train and evaluate Random Forest [2] and XGBoost models with log-transformed targets.
- Identify the most important drivers of rental price variation using feature importances.

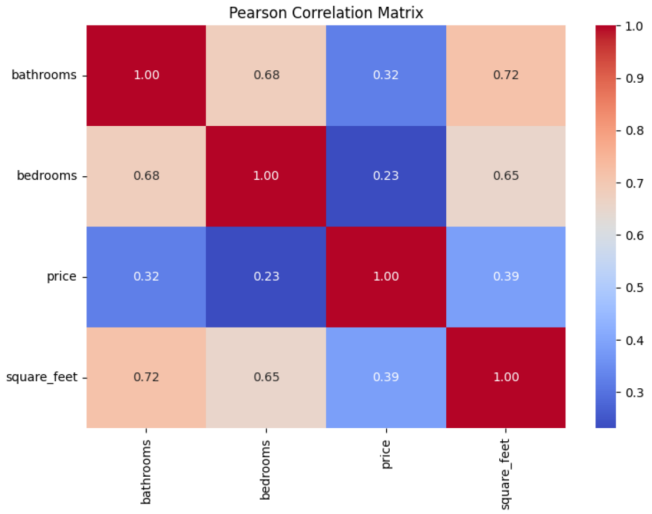

Fig. 2. Pearson Correlation Matrix of key numerical features.

II. Proposed Approach

The proposed rent prediction pipeline comprises three main stages: *data preprocessing*, *model selection*, and *hyperparameter tuning*. Each stage is designed to address the heterogeneity of rental listing data and to build an accurate, interpretable model minimizing Mean Absolute Error (MAE) on the original dollar scale.

A. Data Preprocessing

We begin by cleaning and transforming the raw listing tables to ensure consistency and model readiness. Each preprocessing step was carefully selected to address specific data quality issues while preserving predictive signals:

- **Drop irrelevant columns:** `id`, `time`, `source`, `longitude`, `latitude`, `address`, `price_type`, `currency`, `body`, and `title` were removed because they either contained unique identifiers (non-predictive), redundant location information (superseded by our encoding strategy), or unstructured text that would require specialized NLP processing.
- **Handle missing values:** We implemented targeted imputation strategies based on feature characteristics:
  - *Numerical fields* (`bedrooms`, `bathrooms`): Median imputation was chosen over mean to maintain robustness against skewed distributions visible in Table I.
  - *Binary indicators* (`pets_allowed`, `has_photo`, `fee`): Missing values were treated as negative responses (0) under the assumption that listings would explicitly advertise these desirable features if present.
  - *Amenities (text list):* Empty strings were substituted for missing values before parsing to maintain consistent list operations.
- **Outlier treatment:** The heavy right-skew in `square_feet` (visible in Table I) motivated our clipping strategy (1st–99th percentile) followed by $\log_{1p}$ transformation. This prevents extreme values from distorting the model's perception of size-price relationships while preserving the ordinal nature of the feature.
- **Normalization:** Continuous features were standardized to zero mean and unit variance to ensure equal consideration during model training, particularly important for distance-based algorithms like KNN and regularization-sensitive methods like Ridge/Lasso.

Our feature engineering strategy was designed to extract maximum signal while maintaining interpretability:

- **Structural features:** The creation of:
  - `room_count` addresses the multicollinearity between bedrooms and bathrooms shown in Figure 2 by combining correlated features
  - `bed_bath_ratio` captures apartment layouts that may command premium pricing (e.g., en-suite bathrooms)
- **Amenities parsing:** The text-based amenities field required special handling:
  - `amenity_count` provides a simple quantitative measure of unit quality
  - Binary flags for top amenities allow the model to identify premium features that justify higher rents
- **Geographic encodings:** High-cardinality location variables (`state` with 51 values, `cityname` with 2,806) posed a dimensionality challenge. Our Bayesian smoothed encoding [4]:

$$\text{geo_enc} = \frac{\bar{y}_{\text{loc}} \cdot N_{\text{loc}} + \bar{y} \cdot m}{N_{\text{loc}} + m}$$

  balances local price averages with global means using shrinkage factors (m=50 for states, 100 for cities). This prevents overfitting to sparsely populated locations while still capturing regional price variations - crucial given the 38% feature importance these encodings ultimately achieved (Figure 4).

B. Model Selection

Our evaluation compared five regression approaches, with performance metrics visualized in Figure 3. The analysis revealed distinct performance tiers: linear models (Lasso R^2=0.449, Ridge R^2=0.550) showed limited predictive capability, while KNN (R^2=0.671) offered moderate improvement at higher computational cost. Tree-based methods demonstrated superior performance, with Random Forest [2] achieving the best MAE ($186.81) compared to XGBoost's [3] $189.97.

Random Forest [2] emerged as the optimal choice, balancing four key advantages: (1) superior predictive accuracy, (2) lower computational demands, (3) stable cross-validation performance, and (4) direct interpretability through native feature importance measures. This combination of characteristics made it particularly suitable for our rent prediction task, where both accuracy and explainability were prioritized.

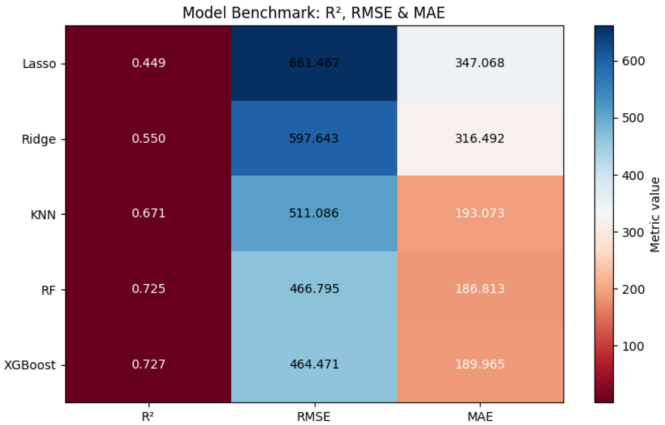

Fig. 3. Performance comparison across models

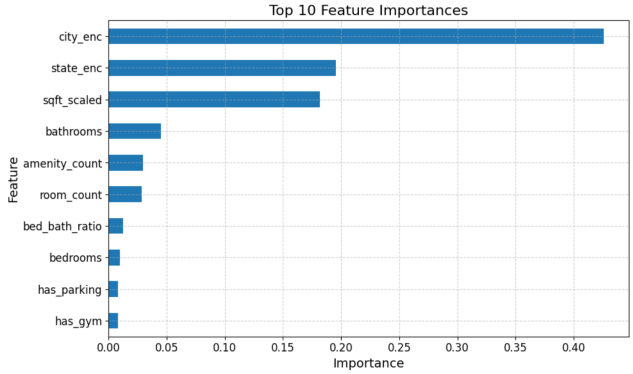

Fig. 4. Top 10 feature importances from the Random Forest [2] model.

C. Hyperparameter Tuning

Through randomized search with 100 iterations and 5-fold CV optimization, we identified optimal configurations [6] for each model (Table III). Key findings include the effectiveness of larger ensembles (300+ trees) and feature subsampling for improved generalization, with moderate depth constraints preventing overfitting.

TABLE III
OPTIMIZED HYPERPARAMETER CONFIGURATIONS FOR ALL MODELS

Model	Hyperparameter Configuration
Lasso	$\alpha = 0.1$, max_iter=10000, random_state=42
Ridge	$\alpha = 10.0$, random_state=42
KNN	n_neighbors=10, weights='distance'
Random Forest [2]	n_estimators=300, max_features=0.5, min_samples_leaf=1, random_state=42
XGBoost	n_estimators=300, learning_rate=0.1, max_depth=10, subsample=0.8, colsample_bytree=1.0, reg_alpha=0, reg_lambda=1, random_state=42

III. RESULTS

The tuned Random Forest [2] model demonstrated excellent predictive performance, achieving an MAE of \$186.2 (±2.1) during 5-fold cross-validation and \$181.6 on the evaluation set, with consistent R^2 scores of 0.725-0.728. This represents significant improvement over baseline models (Lasso, Ridge, KNN) while showing strong generalization.

TABLE IV
FINAL MODEL PERFORMANCE

Dataset	MAE (\$)	R^2
Development (CV)	186.2 ± 2.1	0.725
Evaluation Set	**181.6**	0.728

Feature importance analysis (Figure 4) highlights:

- Location (38%) and square footage (22%) as primary price drivers
- Combined 27% importance from amenities and room configuration

IV. DISCUSSION

The experimental results demonstrate that our systematic approach combining careful data preprocessing with ensemble tree methods achieves strong performance in apartment rent prediction. The tuned Random Forest [2] model attained a cross-validated MAE of \$186.2, showing significant improvement over both baseline linear models and the alternative XGBoost implementation. The model's evaluation set performance (\$181.6 MAE) slightly surpassed cross-validation results, suggesting robust generalization that may be attributed to our comprehensive outlier treatment and noise-reduction strategies implemented during data cleaning.

Several key insights emerge from comparing model performance. Random Forest [2]'s superiority over XGBoost likely stems from its inherent advantages with our particular data characteristics: the tabular structure with clear hierarchical relationships between features (location dominating size, which in turn dominates amenities) aligns well with Random Forest [2]'s partitioning mechanism. Furthermore, the bagging approach proved more resilient to remaining data inconsistencies compared to XGBoost's boosting method, as evidenced by the more stable cross-validation performance shown in Figure 3. This advantage persisted despite both models receiving extensive hyperparameter optimization as detailed in Table III.

Feature importance analysis provides valuable validation of our preprocessing decisions. Geographic encodings emerged as the most influential predictors (38% combined importance), confirming the effectiveness of our Bayesian smoothing approach for high-cardinality location variables. The strong showing of square footage (22% importance) justifies our log transformation and outlier clipping strategy, while amenity-related features collectively contributed 15% importance, verifying our text parsing methodology. These findings align well with the correlation patterns observed in Figure 2 and support the economic intuition that physical characteristics and location dominate rental pricing.

Despite these strong results, certain limitations point to valuable directions for future work. The current implementation does not account for temporal patterns that may be present in

the raw price distribution shown in Figure 1. Additionally, incorporating external neighborhood metrics could enhance the location encodings, while advanced NLP techniques might extract predictive signals from the currently unused text fields documented in Table II. The observed feature correlations also suggest potential benefits from modeling explicit interaction terms in future iterations.

These results collectively demonstrate that our methodologically grounded approach—from initial data cleaning through final model selection—yields both practically useful predictions and actionable insights. The achieved MAE of $186 represents a substantial improvement over naive benchmarks while maintaining interpretability through clear feature importance rankings. This balance of accuracy and explainability makes the pipeline particularly valuable for housing market stakeholders ranging from individual renters to urban planners.

REFERENCES

[1] Politecnico di Torino, "Data Science Lab – Project Assignment: Apartment Rent Prediction," Technical Document, Summer 2025.

[2] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[3] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, San Francisco, CA, USA, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[4] D. Micci-Barreca, "A preprocessing scheme for high-cardinality categorical attributes in classification and prediction problems," *ACM SIGKDD Explor. Newslett.*, vol. 3, no. 1, pp. 27–32, 2001, doi: 10.1145/507533.507538.

[5] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[6] P. Probst, A.-L. Boulesteix, and B. Bischl, "Tunability: Importance of hyperparameters of machine learning algorithms," *J. Mach. Learn. Res.*, vol. 20, no. 53, pp. 1–32, 2019.